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From this assignment, I learned how different networking technologies work together in a complete network. I learned how to configure VLANs and use router-on-a-stick with subinterfaces to provide communication between different VLANs. I also learned how to configure DHCP on the router so that devices can automatically receive IP addresses and default gateways. On the switch, I practiced configuring trunk and access ports, Rapid PVST+, a management VLAN, and a blackhole VLAN for unused ports. I also learned how to configure a wireless router, including its Internet connection, LAN network, hidden SSID, and WPA2-Personal security. Finally, I learned how to connect a wireless guest device and verify its IP addressing and connectivity. This challenge helped me understand how these configurations work together in a real network.

Cisco Packet Tracer - /home/abrish/Desktop/IBT/Cybersecurity/module 2/Labs/day 5/2.0 Skills Integration Challenge.pka - Guest - 2026-08-17 18:40:03

File Edit Options View Tools Extensions Window Help

PT Activity: 00:24:56

Packet Tracer – Skills Integration Challenge

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	209.165.200.1	255.255.255.224	N/A
	G0/1.10	172.17.10.1	255.255.255.0	N/A
	G0/1.20	172.17.20.1	255.255.255.0	N/A
	G0/1.88	172.17.88.1	255.255.255.0	N/A
	G0/1.99	172.17.99.1	255.255.255.0	N/A
S2	VLAN 99	172.17.99.32	255.255.255.0	172.17.99.1
WRS	Internet	DHCP Assigned	DHCP Assigned	DHCP Assigned
	LAN	172.17.40.1	255.255.255.0	N/A

Scenario

In this challenge activity, you will configure VLANs and inter-VLAN routing, DHCP, and Rapid PVST+. You will also be required to configure a Linksys router for wireless connectivity with wireless security. At the end of the activity, the PCs will not be able to ping each other but should be able to ping the outside host.

Requirements

R1 Configurations

- Enable and configure the subinterfaces with the following requirements:
 - Configure IP addressing for the subinterfaces according to the Addressing Table.
 - Configure the appropriate dot1Q encapsulation.
 - Configure VLAN 99 as the native VLAN.
- Configure DHCP pools for VLAN 10, 20 and 88 with the following requirements:
 - Next the DHCP pools for VLAN 10, 20 and 88.

Time Elapsed: 00:24:56 Completion: 100/100

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